

Akram Mohamed Hassan

AI Engineer

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SUMMARY

Recent Computer Science graduate looking to start my career in Artificial Intelligence. I have a genuine interest in both the research side and the practical side of AI, and I enjoy learning how intelligent systems are built and how they work. While I am still early in my journey, I am motivated, eager to learn, and ready to put in the work to grow in this field. I am looking for an opportunity where I can contribute, develop my skills, and work toward building real AI systems.

PROFESSIONAL EXPERIENCE

Agentic AI Developer Trainee

01/2026 – 04/2026

National Technology Institute (NTI) Hire-Ready R13

Smart Village , Egypt

Built a strong foundation in LLMs, transformer architectures, and NLP pipelines, with practical exposure to prompt engineering, memory systems, vector databases, and RAG. Gained hands-on experience building multi-agent systems using LangChain, with knowledge spanning reasoning, planning, external integrations, and end-to-end deployment and monitoring of AI systems.

Enhanced soft skills of professional development in business communication, leadership, project management, and teamwork, culminating in a capstone project presentation demonstrating cross-functional collaboration

PROJECTS

Code-M8: Autonomous Coding Agent

- Built an autonomous coding agent using LLM-driven planning with structured reasoning and tool usage.
- Designed a multi-agent architecture with specialized sub-agents for planning, execution, and validation.
- Implemented context engineering and context isolation to manage long-horizon tasks and prevent context pollution.
- Integrated memory systems and a GraphRAG pipeline for retrieval, knowledge grounding, and improved response accuracy.
- Developed robust tool-calling pipelines for code execution, debugging, and external API interaction.
- Enabled code execution and sandboxed environments for safe runtime evaluation and debugging.
- Applied techniques in task decomposition, reflection, and iterative self-improvement for complex problem solving.
- Tools Used: LangChain, LangGraph, DeepAgents SDK, Vector Databases, Neo4j.

TalkToPDF

Hybrid RAG-based Document QA System Built an end-to-end Retrieval-Augmented Generation (RAG) system to enable conversational querying over PDF documents using **LLaMA 3 via Groq**.

- Implemented hybrid retrieval combining semantic search (ChromaDB) and keyword-based BM25
- Integrated re-ranking with FlashRank to improve answer relevance
- Developed conversational memory for multi-turn question answering
- Enabled source-grounded responses with page-level citations
- Designed scalable pipeline for multi-document ingestion

Tech Stack: Python, Streamlit, LangChain, ChromaDB, Sentence-Transformers, Rank-BM25, FlashRank, Groq

Sentiment Analysis

Sentiment Analysis on IMDB Reviews using LSTM Developed a deep learning model for binary sentiment classification on the **IMDB Movie Reviews Dataset** using LSTM networks in **TensorFlow**.

- Preprocessed and tokenized text data, including padding and vocabulary encoding
- Designed and trained an LSTM-based architecture for sequence modeling
- Achieved robust performance on unseen data through proper validation and tuning
- Built an interactive web interface for real-time sentiment prediction
- Deployed the model as a lightweight web app using **Streamlit**

Tools: Python, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, NLTK, Streamlit

SKILLS

Programming & Core

- Python,C++, SQL, OOP, Data Structures & Algorithms
- Git, GitHub

Machine Learning & Deep Learning

- Supervised & Unsupervised Learning
- Neural Networks, CNNs, RNNs, LSTM
- Model Evaluation, Hyperparameter Tuning

LLMs & Generative AI

- Retrieval-Augmented Generation (RAG)
- Prompt Engineering , Fine Tuning

Frameworks & Libraries

- TensorFlow, PyTorch
- Scikit-learn, Pandas, NumPy
- LangChain, Hugging Face

SOFT SKILLS

- Problem Solving • Communication Skills • Teamwork • Adaptability • Project Management

EDUCATION

B.S. in Computers and Information

Faculty of Computers and Artificial Intelligence, Cairo University
GPA: 2.94

2020 – 2024
Giza,Egypt

CERTIFICATIONS

Data Scientist Professional with Python

DataCamp

2023

LANGUAGES

Arabic



English

